

5. Equilibrium of a Rigid Body

Thursday, September 12, 2024 9:37 AM

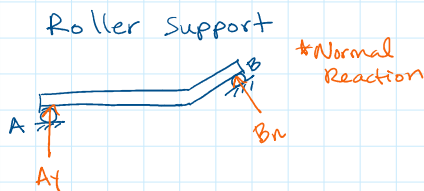
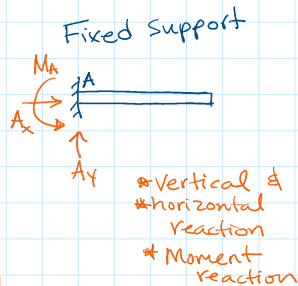
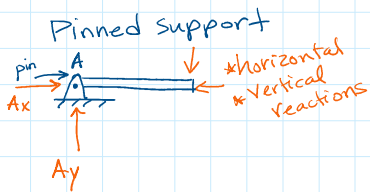
- * When a body is in a state of Equilibrium its acceleration is zero.
 - * Rigid Body - a body that does not deform (shape is not altered) due to forces being exerted on it.
 - * Sum of Forces equals zero $\sum F = 0$
 - * Sum of Moments equals zero $\sum M = 0$
- New $\rightarrow$

In 2D $\rightarrow \sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_o = 0$
 $\rightarrow \sum F_n = 0 \quad \sum F_t = 0 \quad \sum M_o = 0$

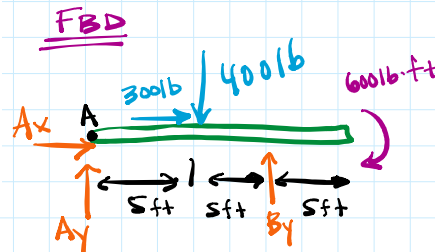
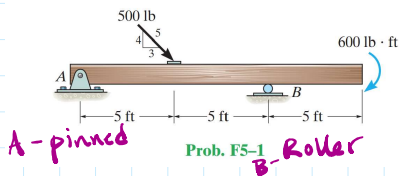


$$F_s = k(l - l_0) \text{ - Spring Force}$$

Types of Supports



F5-1. Determine the horizontal and vertical components of reaction at the supports. Neglect the thickness of the beam.



$$\sum \vec{F} = 0 \quad \sum M_A = 0$$

$$\rightarrow \sum F_x = Ax + 300 = 0$$

$$Ax = -300 \text{ lb}$$

$$Ax = 300 \text{ lb} \leftarrow$$

$$\uparrow \sum F_y = Ay + By - 400 = 0$$

$$Ay + By = 400$$

$$Ay = 140 \text{ lb} \uparrow$$

$$\curvearrowright \sum M_A = -600 + 10By - 5(400) = 0$$

$$10By = 2600$$

$$By = 260 \text{ lb} \uparrow$$

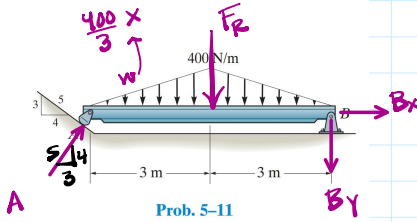
$$\sum \mathcal{M}_A = -600 + 10B_y - 3(400) = 0$$

$$10B_y = 2600$$

$$B_y = 260 \text{ lb } \uparrow$$

5-11. Determine the reactions at the supports.

A - roller
B - pinned



Prob. 5-11

$$F_R = \frac{1}{2}bh = \frac{1}{2}(6\text{m})(400\text{N/m})$$

$$F_R = 1200 \text{ N}$$

$$A_x = \frac{3}{5}A$$

$$A_y = \frac{4}{5}A$$

$$\sum \mathcal{M}_A = -3(F_R) - 6B_y = 0$$

$$B_y = -\frac{3}{6}F_R = -\frac{1}{2}(1200) = -600 \text{ N}$$

$$B_y = 600 \text{ N } \uparrow$$

$$\sum F_x = 0$$

$$A_x + B_x = 0$$

$$A_x = -B_x$$

$$\sum F_y = 0$$

$$A_y - F_R - B_y = 0$$

$$A_y = 1200 + B_y$$

$$\frac{4}{5}A = 1200 + B_y$$

$$B_x = -\frac{3}{5}(750)$$

$$B_x = -450 \text{ N}$$

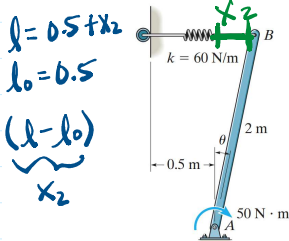
$$B_x = 450 \text{ N } \leftarrow$$

$$\frac{4}{5}A = 1200 - 600$$

$$A = \frac{5}{4}(600)$$

$$A = 750 \text{ N}$$

*5-44. The 10-kg uniform rod is pinned at end A. If it is also subjected to a couple moment of 50 N·m, determine the smallest angle θ for equilibrium. The spring is unstretched when $\theta = 0$, and has a stiffness of $k = 60 \text{ N/m}$.



Prob. 5-44

$$m = 10 \text{ kg}$$

$$k = 60 \text{ N/m}$$

$$M = 50 \text{ N}\cdot\text{m}$$

$$l_0 = 0.5 \text{ m}$$

$$l_{rod} = 2 \text{ m}$$

Pinned @ A

unknown A_x & A_y

$$W = 10 \text{ kg } (9.81 \text{ m/s}^2)$$

$$F_s = k(l - l_0)$$

$$\sum \mathcal{M}_A = 0$$

$$\sum \mathcal{M}_A = -50 - x_1 W + y_2 F_s = 0$$

$$50 = -\sin\theta \cdot W + 2 \cdot \cos\theta F_s$$

$$50 = -\sin\theta \cdot W + 2 \cdot \cos\theta \cdot k \cdot 2 \sin\theta$$

$$50 = -9.81 \sin\theta + 240 \sin\theta \cdot \cos\theta$$

$$5 = -9.81 \sin\theta + 24 \sin\theta \cdot \cos\theta$$

$$y = 5$$

$$y = -9.81 \sin\theta + 24 \sin\theta \cos\theta$$

$$\sin\theta = \frac{x_1}{1}$$

$$x_1 = \sin\theta$$

$$\cos\theta = \frac{y_2}{2}$$

$$\sin\theta = \frac{x_2}{2}$$

$$x_2 = 2 \cdot \sin\theta$$

$$y_2 = 2 \cdot \cos\theta$$

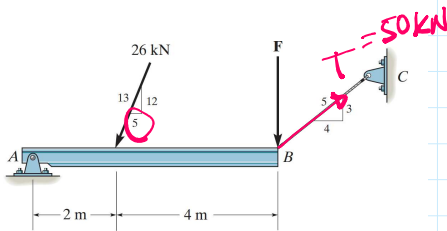
$$\cos \theta = \frac{y_2}{2}$$

$$(y_2 = 2 \cdot \cos \theta)$$

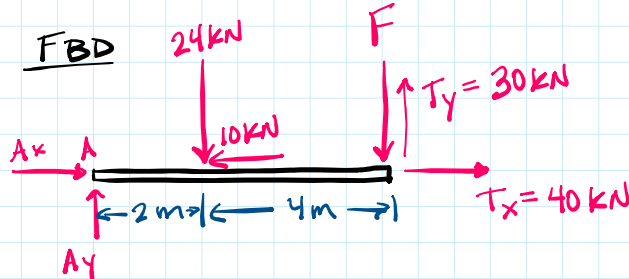
solve graphically

$$\theta = 0.429 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right) = 24.58^\circ$$

*5-48. If rope BC will fail when the tension becomes 50 kN, determine the greatest vertical load F that can be applied to the beam at B. What is the magnitude of the reaction at A for this loading? Neglect the thickness of the beam.



Probs. 5-47/48



$$\sum M_A = -2(24) - 6(F) + 6(30) = 0$$

$$-48 + 180 = 6F$$

$$\frac{132}{6} = \frac{6F}{6}$$

$$F = 22 \text{ kN} \downarrow$$

$$F < 22 \text{ kN}$$

$$\sum F_x = Ax - 10 + 40 = 0$$

$$Ax = -30 \text{ kN}$$

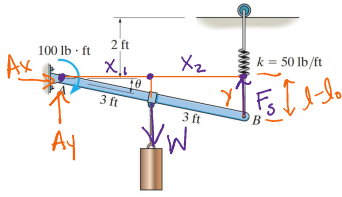
$$Ax = 30 \text{ kN} \leftarrow$$

$$\sum F_y = Ay - 24 - F + 30 = 0$$

$$Ay = 24 + 22 - 30$$

$$Ay = 16 \text{ kN} \uparrow$$

5-59. The rod supports a weight of 200 lb and is pinned at its end A. If it is also subjected to a couple moment of 100 lb · ft, determine the angle θ for equilibrium. The spring has an unstretched length of 2 ft and a stiffness of $k = 50$ lb/ft.



Prob. 5-59

$$W_M = 200 \text{ lb}$$

Pinned @ A, A_x & A_y

$$M = 100 \text{ lb} \cdot \text{ft}$$

$$l_0 = 2 \text{ ft}, k = 50 \text{ lb/ft}, l_{\text{rod}} = 6 \text{ ft}$$

$$\sum M_A = -x_1 W + (x_1 + x_2) F_s - 100 = 0$$

$$x_1 = \frac{l_r \cos \theta}{2}$$

$$x_1 + x_2 = l_r \cos \theta$$

$$F_s = k (\underbrace{l - l_0}_{\text{amount of stretch}}) = k y$$

$$F_s = k \cdot l_r \cdot \sin \theta$$

$$\sum M_A = -\frac{l_r \cos \theta}{2} \cdot W + (l_r \cos \theta) k \cdot l_r \sin \theta - 100 = 0$$

$$-\frac{6}{2} (200) \cos \theta + (6 \cdot \cos \theta) (50) (6) \sin \theta - 100 = 0$$

$$-600 \cos \theta + 1800 \cos \theta \sin \theta - 100 = 0$$

$$-6 \cos \theta + 18 \cos \theta \sin \theta - 1 = 0$$

$$18 \cos \theta \sin \theta = 1 + 6 \cos \theta$$

$$3 \cos \theta \sin \theta = \frac{1}{6} + \cos \theta$$

graph with desmos

graph with desmos.

$$3 \cos x \sin x = \frac{1}{6} + \cos x$$

find point of intersection

(x, y)

answer in radians.